- ## ANUSH MINOR PROJECT –

```
# Essential libraries for data handling, visualization, and modeling
import pandas as pd
import numpy as np

import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
   accuracy_score, precision_score, recall_score,
   confusion_matrix, classification_report
)
# Upload 'advertising.csv' using the left panel in Google Colab before running this cell.

df = pd.read_csv("advertising.csv")

print("Dataset loaded. Preview of first rows:")
df.head()
```

Dataset loaded. Preview of first rows:

	Daily Time Spent on Site	Age	Area Income	Daily Internet Usage	Ad Topic Line	City	Male	Country	Timestamp	Clicked on Ad
0	68.95	35	61833.90	256.09	Cloned 5thgeneration orchestration	Wrightburgh	0	Tunisia	2016-03-27 00:53:11	0
1	80.23	31	68441.85	193.77	Monitored national standardization	West Jodi	1	Nauru	2016-04-04 01:39:02	0
2	69.47	26	59785.94	236.50	Organic bottom-line service-desk	Davidton	0	San Marino	2016-03-13 20:35:42	0
3	74.15	29	54806.18	245.89	Triple-buffered reciprocal time-frame	West Terrifurt	1	Italy	2016-01-10 02:31:19	0
4	68.37	35	73889.99	225.58	Robust logistical utilization	South Manuel	0	Iceland	2016-06-03 03:36:18	0

```
print("\nGeneral Dataset Information:\n")
df.info()

print("\nStatistical Overview:\n")
df.describe()
```

```
General Dataset Information:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Daily Time Spent on Site  1000 non-null   float64
 1   Age                       1000 non-null   int64
 2   Area Income               1000 non-null   float64
 3   Daily Internet Usage      1000 non-null   float64
 4   Ad Topic Line             1000 non-null   object
 5   City                      1000 non-null   object
 6   Male                      1000 non-null   int64
 7   Country                   1000 non-null   object
 8   Timestamp                 1000 non-null   object
 9   Clicked on Ad             1000 non-null   int64
dtypes: float64(3), int64(3), object(4)
memory usage: 78.3+ KB
```

Statistical Overview:

	Daily Time Spent on Site	Age	Area Income	Daily Internet Usage	Male	Clicked on Ad
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000
mean	65.000200	36.009000	55000.000080	180.000100	0.481000	0.50000
std	15.853615	8.785562	13414.634022	43.902339	0.499889	0.50025
min	32.600000	19.000000	13996.500000	104.780000	0.000000	0.00000
25%	51.360000	29.000000	47031.802500	138.830000	0.000000	0.00000
50%	68.215000	35.000000	57012.300000	183.130000	0.000000	0.50000
75%	78.547500	42.000000	65470.635000	218.792500	1.000000	1.00000
max	91.430000	61.000000	79484.800000	269.960000	1.000000	1.00000

---

print("Missing values per column:\n")
print(df.isnull().sum())

```
Missing values per column:

Daily Time Spent on Site    0
Age                         0
Area Income                 0
Daily Internet Usage        0
Ad Topic Line               0
City                        0
Male                        0
Country                     0
Timestamp                   0
Clicked on Ad               0
dtype: int64
```

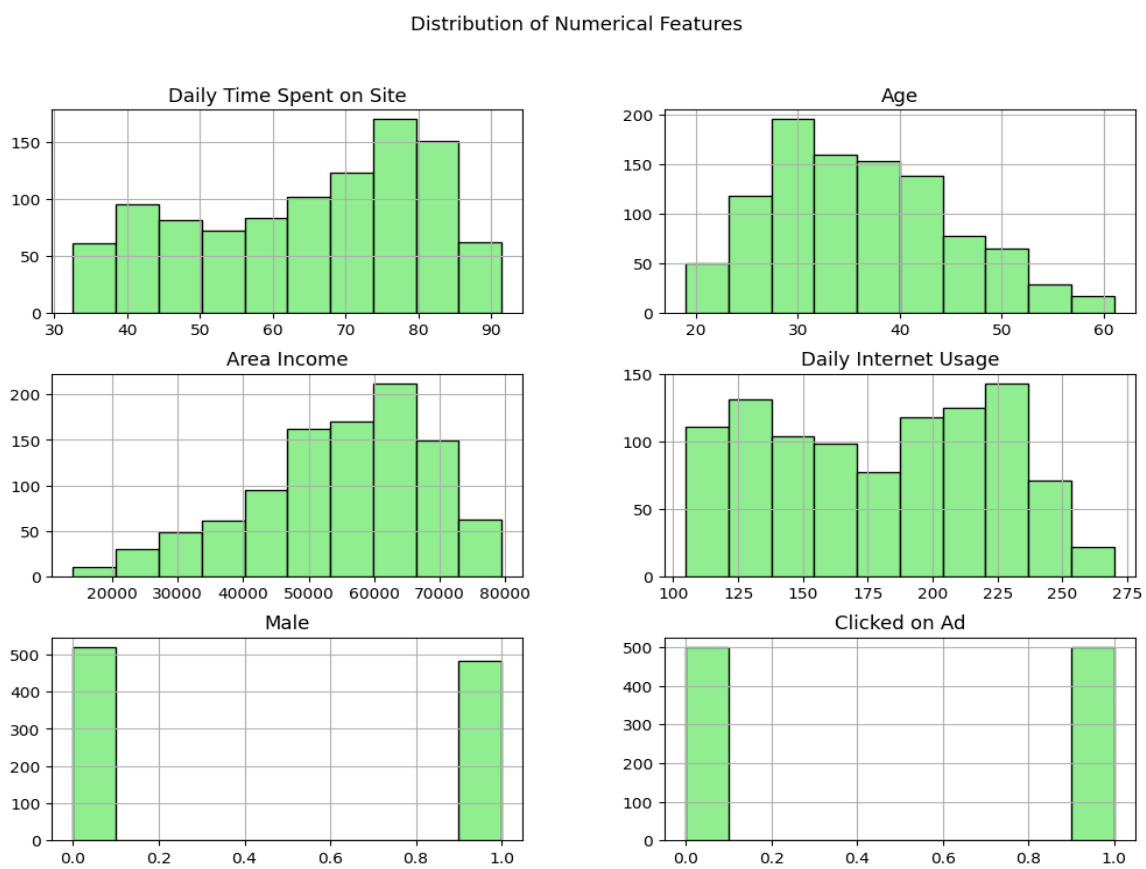
---

print("Columns present in the dataset:")
print(df.columns.tolist())

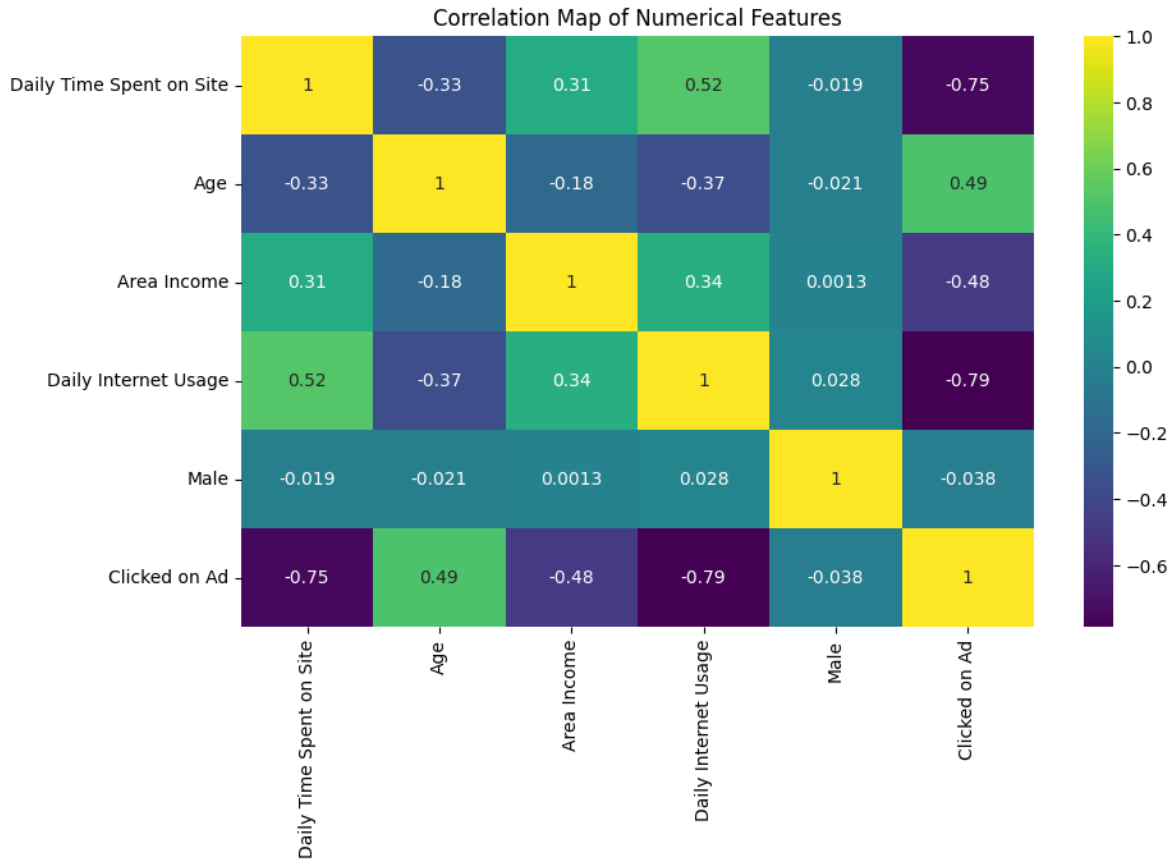
```
Columns present in the dataset:
['Daily Time Spent on Site', 'Age', 'Area Income', 'Daily Internet Usage', 'Ad Topic Line', 'City', 'Male', 'Country', 'Timestamp', 'Clicked on Ad']
```

---

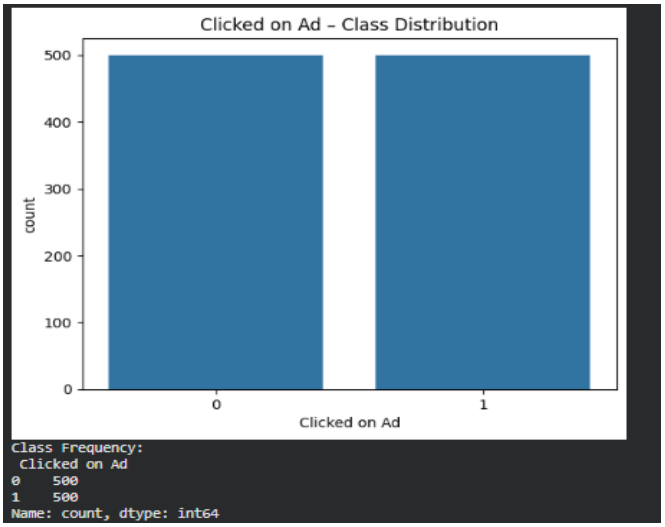
df.hist(figsize=(12, 9), color="lightgreen", edgecolor="black")
plt.suptitle("Distribution of Numerical Features")
plt.show()



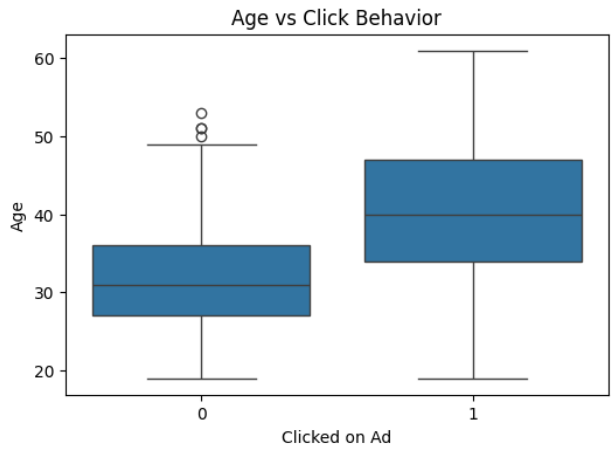
```
plt.figure(figsize=(10, 6))
# Select only numerical columns for correlation calculation
numeric_df = df.select_dtypes(include=[np.number])
sns.heatmap(numeric_df.corr(), annot=True, cmap="viridis")
plt.title("Correlation Map of Numerical Features")
plt.show()
```

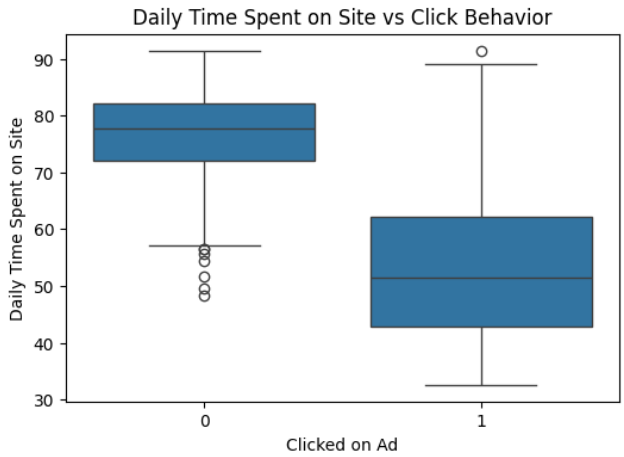


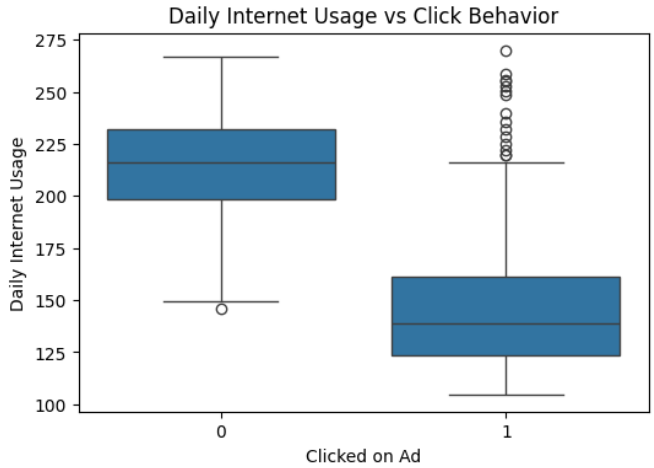
```
sns.countplot(x=df["Clicked on Ad"])
plt.title("Clicked on Ad – Class Distribution")
plt.show()
print("Class Frequency:\n", df["Clicked on Ad"].value_counts())
```



```
# Boxplots to see how features vary across target categories

features_to_plot = ["Age", "Daily Time Spent on Site", "Daily Internet Usage"]

for feature in features_to_plot:
    plt.figure(figsize=(6,4))
    sns.boxplot(x="Clicked on Ad", y=feature, data=df)
    plt.title(f"{feature} vs Click Behavior")
    plt.show()
```







```
X = df[["Daily Time Spent on Site",
      "Age",
      "Area Income",
      "Daily Internet Usage",
      "Male"]]

y = df["Clicked on Ad"]

print("Selected Feature Matrix (X):")
print(X.head())
```

```
Selected Feature Matrix (X):
   Daily Time Spent on Site  Age  Area Income  Daily Internet Usage  Male
0                     68.95   35     61833.90                256.09     0
1                     80.23   31     68441.85                193.77     1
2                     69.47   26     59785.94                236.50     0
3                     74.15   29     54806.18                245.89     1
4                     68.37   35     73889.99                225.58     0
```

```
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.30, random_state=10)

print("Training Set Shape:", X_train.shape)
print("Test Set Shape:", X_test.shape)
```

```
Training Set Shape: (700, 5)
Test Set Shape: (300, 5)
```

```
log_model = LogisticRegression()

log_model.fit(X_train, y_train)

print("Model has been fitted to the training data.")
```

```
Model has been fitted to the training data.
```

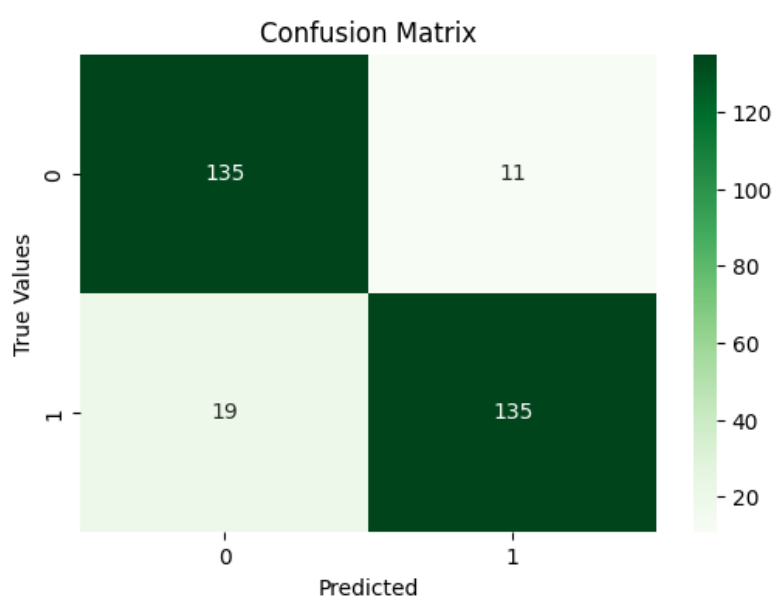
```
y_pred = log_model.predict(X_test)

print("Sample predictions:", y_pred[:8])
```

```
Sample predictions: [1 1 0 0 1 1 0 0]
```

```
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred)
rec = recall_score(y_test, y_pred)

print("Performance Metrics:")
print("---------------------")
print(f"Accuracy : {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall   : {rec:.4f}")

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

```
Performance Metrics:
---------------------
Accuracy : 0.9000
Precision: 0.9247
Recall   : 0.8766

Classification Report:

              precision    recall  f1-score   support

           0       0.88      0.92      0.90       146
           1       0.92      0.88      0.90       154

    accuracy                           0.90       300
   macro avg       0.90      0.90      0.90       300
weighted avg       0.90      0.90      0.90       300
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, cmap="Greens", fmt="d")
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("True Values")
plt.show()
```



```
----------------------------------------------------------------------


print("""
 KEY TAKEAWAYS FROM THE MODEL
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1. Logistic Regression turned out to be suitable for predicting user
   click behavior since the output is binary.

2. Features such as Daily Time Spent on Site, Age, and Internet Usage
   show strong patterns with respect to ad-click likelihood.

3. Model performance was solid, with high accuracy, precision,
   and recall—indicating reliable predictions.

4. The model can assist ConnectSphere Digital in filtering out users
   who are unlikely to click, helping optimize marketing budgets.

5. This translates to better CTR, reduced wastage, and more efficient
   campaign performance.

 Overall, the project successfully builds a predictive system that supports
data-driven decision-making for online advertising.
""")
```